Re-Designing Everyday Products for Sustainability

Subtitle: Expiry-Sensing Food Storage Box

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Problem Statement

Food spoilage is a common issue faced in daily life. Many people are unable to identify whether food is safe to consume, which leads to health risks and unnecessary waste.

Key Issues:

- No clear indication of food freshness
- Consumption of spoiled food can cause illness
- Large amount of food is wasted due to uncertainty
- Lack of affordable smart solutions

SGD Connection

This innovation directly supports Sustainable Development Goal 12, which focuses on reducing waste and promoting efficient use of resources.

Contribution:

Encourages responsible consumption

Minimizes food wastage

Promotes sustainable living practices

Proposed Solution

The Expiry-Sensing Food Storage Box is designed to give users a clear visual indication of food freshness.

Core Idea:

- Detects spoilage gases such as ammonia
- Uses a color-changing indicator
- Provides instant visual feedback

👉If food spoils → color changes → user is alerted

How it Works

The system works using a simple chemical principle. When food spoils, it releases gases that change the pH level inside the container.

Working Mechanism:

- Spoilage releases gases (ammonia, sulfur compounds)
- These gases affect the internal pH
- A pH-sensitive strip changes color

Example:

- Fresh food → Blue/Green
- Spoiled food → Yellow/Red

Unique Features

This product stands out because it avoids complex technology and focuses on accessibility.

Key Features:

- No mobile app required
- No electricity needed
- Low-cost and easy to manufacture
- Suitable for rural and urban areas
- Eco-friendly design

Design & Components

The design is simple but efficient, making it easy to use and produce.

Main Components:

- Airtight food container
- Gas-permeable indicator section
- pH-sensitive strip
- Transparent viewing window
- The design ensures that the indicator is visible without opening the container.

Advantages

This innovation provides benefits across multiple areas.

Health:

Prevents consumption of spoiled food

Economic:

Reduces unnecessary food wastage

Saves money

Environmental:

Reduces landfill waste

Lowers harmful gas emissions

Applications

The product is versatile and can be used in many real-life situations.

Use Cases:

- Household food storage
- School and office lunchboxes
- Restaurants and catering services
- Rural areas without electricity
- Travel and outdoor use

Engineer’s Role

Engineers are responsible for converting this idea into a practical, safe, and scalable product.

Roles:

- Selecting safe and accurate chemical indicators
- Designing durable and user-friendly containers
- Ensuring affordability
- Testing performance and reliability
- Improving sustainability

Future Improvements

This concept has potential for further development.

Enhancements:

- Detection of multiple gases
- Replaceable indicator strips
- Use of biodegradable materials
- Optional smart/digital integration

The Expiry-Sensing Food Storage Box is a simple yet powerful innovation that addresses both food safety and sustainability.

It highlights that:

- Smart solutions can be simple
- Low-cost ideas can have high impact
- Engineering plays a vital role in sustainable development

“ Design smarter. Design inclusive. Design sustainable.”

Thank You!!